Quantum computing for effective nuclear lattice model

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I. INTRODUCTION

The effective lattice formulation provides a useful framework for describing low-energy nuclear systems. [More introduction on the lattice model and the classical methods, discussing their strength and limitations].

Quantum computing provides a potential route to address some of the computational challenges faced by classical methods for lattice nuclear Hamiltonians. A quantum device can represent many-body wave functions in a way that scales more naturally with system size, and may help to overcome the computational limits of classical approaches for strongly correlated systems. For many classical approaches, such as Monte Carlo–based methods, the computational cost grows rapidly with the system size. In lattice formulations, the lattice size L controls the spatial resolution and the finite-volume error. In many settings, the lattice size required for a target accuracy does not increase rapidly with the number of particles n, which motivates exploring quantum algorithms for few- and intermediate-body calculations.

To solve the lattice model on a quantum computer, one must choose an encoding that maps fermionic basis states and operators to qubits. A standard choice is the Jordan–Wigner (JW) transformation, which maps each single-particle state to one qubit. For the lattice model considered here, with four spin–isospin flavors per site, the JW register requires $4L^3$ qubits. This scaling is compatible with fault-tolerant devices but becomes demanding on near-term hardware at moderate L. An alternative is to first reduce the effective Hilbert space using symmetries and then encode only the reduced basis. Gray code encoding follows this approach and maps the symmetry-adapted basis states of the reduced Hilbert space directly to a compact qubit register.

From the algorithmic perspective, methods such as quantum phase estimation and quantum signal processing offer rigorous guarantees in principle, but they typically require deep circuits and large qubit counts. In the noisy intermediate-scale quantum (NISQ) regime, variational methods are more practical. In particular, the variational quantum eigensolver (VQE) uses parameterized circuits with relatively shallow depth, and has been widely adopted to solve quantum many-body problems across quantum chemistry and condensed-matter physics.

Despite the potential of quantum computing for nuclear lattice models, there have been few efforts to study solving this model with quantum algorithms. In this work, we present a systematic study of solving a nuclear lattice Hamiltonian with VQE. We first construct the lattice Hamiltonian in a second-quantized form. We provide a detailed description and comparison between the JW encoding and Gray code encoding in terms of construction and quantum resource requirements, including the number of qubits and the number of Hamiltonian terms (the latter determines the measurement cost). For the cases considered in this work ($n = 2, 3, 4$) and lattice sizes $L = 2$–6, we adopt Gray code encoding, which directly encodes the basis states of the reduced Hilbert space. To reduce quantum resources, we use spatial symmetries and parity ($\mathbb{Z}_2$) symmetries associated with conserved quantities to reduce the effective Hilbert space. This makes it possible to study all cases with a small number of qubits. We then present numerical results for these systems. As L increases, the computed ground-state energies move toward the corresponding experimental binding energies. We also report the VQE convergence behavior as a function of n and L, and observe that larger n and L typically require more iterations to converge.

The paper is organized as follows. Section II introduces the lattice Hamiltonian and the VQE framework used in this work, including the two encoding methods. Section III describes the symmetry reductions used to reduce the effective Hilbert space and qubit resources. Section IV presents numerical results. Section V concludes the paper.

II. THE NUCLEAR LATTICE MODEL

(Perhaps add more detailed introduction on the lattice model.) Nuclear lattice effective field theory (NLEFT) is a powerful framework tailored for *ab initio* nuclear calculations. In this method, nuclear forces derived from chiral effective field theory are discretized and regularized on a cubic spatial lattice. Constructing the lattice Hamiltonian requires defining the many-body Hilbert space in the lattice coordinate representation. This Hamiltonian is obtained through the discretized second quantization of the corresponding continuum Hamiltonian. In accordance with effective field theory principles, the coupling constants explicitly depend on the momentum cutoff imposed by the lattice. This section details the lattice Hamiltonian and the corresponding coupling constants.

A. Lattice Hamiltonian

Throughout this work, we consider a nuclear Hamiltonian for fermions with four spin-isospin degrees of freedom

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on a three-dimensional lattice with periodic boundary conditions. The Hamiltonian is given by:

$$H = \sum_{\alpha=1}^{A} \frac{\mathbf{p}_\alpha^2}{2m} + \sum_{\alpha<\beta}^{A} V_2(\mathbf{r}_\alpha, \mathbf{r}_\beta) + \sum_{\alpha<\beta<\gamma}^{A} V_3(\mathbf{r}_\alpha, \mathbf{r}_\beta, \mathbf{r}_\gamma). \tag{1}$$

The first term represents the kinetic energy, while the second and third terms correspond to the two-body and three-body interaction potentials, respectively; interactions involving more particles are truncated in this model. (Should we discuss why we can do this truncation here?) Here, A and m denote the number and mass of the fermions, respectively. The indices α, β and γ label distinct fermions. The operator $\mathbf{p}_\alpha = -i\nabla_\alpha$ denotes the momentum, and $\mathbf{r}_\alpha$ represents the coordinate of the α-th fermion. The interaction potentials are modeled as contact interactions, defined as $V_2 = c_2\delta(\mathbf{r}_\alpha - \mathbf{r}_\beta)$ and $V_3 = c_3\delta(\mathbf{r}_\alpha - \mathbf{r}_\beta)\delta(\mathbf{r}_\beta - \mathbf{r}_\gamma)$, where c_2 and c_3 are the corresponding coupling constants.

Upon discretizing the coordinates and applying second quantization, Eq. (1) transforms into:

$$H = \sum_{\sigma=0}^{3} \sum_{l,l'=0}^{L^3-1} T_{ll'} \hat{c}_{l,\sigma}^\dagger \hat{c}_{l',\sigma} + c_2 \sum_{\sigma<\sigma'}^{3} \sum_{l=0}^{L^3-1} \hat{n}_{l,\sigma}\hat{n}_{l,\sigma'} + c_3 \sum_{\sigma<\sigma'<\sigma''}^{3} \sum_{l=0}^{L^3-1} \hat{n}_{l,\sigma}\hat{n}_{l,\sigma'}\hat{n}_{l,\sigma''}, \tag{2}$$

where $T_{l,l'}$ represents the hopping coefficient between lattice sites l and l', derived from the kinetic energy, and L^3 denotes the total number of lattice sites. The index $\sigma \in \{0, 1, 2, 3\}$ labels the four distinct spin-isospin states $\{p\uparrow, p\downarrow, n\uparrow, n\downarrow\}$, where p (n) denotes a proton (neutron) and $\uparrow$ ($\downarrow$) indicates spin-up (spin-down). The operators $\hat{c}_{l,\sigma}^\dagger$ and $\hat{c}_{l,\sigma}$ are the creation and annihilation operators, respectively, for a fermion in state σ at site l, while $\hat{n}_{l,\sigma} = \hat{c}_{l,\sigma}^\dagger \hat{c}_{l,\sigma}$ is the corresponding number operator. In this study, we adopt the parameter values $c_2 = -143.202887$ and $c_3 = 107.232753$, with a lattice spacing of $a = 1.160747$ fm. Specifically, these values were determined by fitting the binding energies of the deuteron, triton, and helium-4, calculated using this Hamiltonian on a three-dimensional lattice with a linear size of $L = 6$. Is it appropriate to explain how these numbers come from?

Another thing to check: In this model we consider particle hopping between arbitrary sites. Does it work for very large L?

III. VQE FRAMEWORK FOR NUCLEAR HAMILTONIANS

In this section, we provide a brief introduction to the variational quantum eigensolver (VQE), which is employed to compute the ground-state energy of the nuclear Hamiltonian. The VQE algorithm relies on the Rayleigh-Ritz variational principle. According to this principle, the expectation value of the Hamiltonian H with respect to a parameterized trial state $|\psi(\boldsymbol{\theta})\rangle$ provides an upper bound on the true ground-state energy E_g:

$$E_g \leq \frac{\langle\psi(\boldsymbol{\theta})|H|\psi(\boldsymbol{\theta})\rangle}{\langle\psi(\boldsymbol{\theta})|\psi(\boldsymbol{\theta})\rangle}. \tag{3}$$

A typical VQE workflow consists of the following steps.
Hamiltonian representation: The first step in VQE is to construct the target Hamiltonian in a form suitable for quantum computation.
Encoding: By employing a fermion-to-qubit encoding scheme, such as the Gray code transformation, the fermionic Hamiltonian is mapped to a qubit Hamiltonian $H = \sum_i c_i P_i$. Here, $P_i \in \{I, X, Y, Z\}^{\otimes N}$ represents an N-qubit Pauli string that can be directly measured on a quantum computer.
Ansatz construction: An expressive yet depth-efficient parameterized quantum circuit $U(\boldsymbol{\theta})$ is constructed to prepare the trial state $|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|\phi_0\rangle$, where $|\phi_0\rangle$ is the initial state.
Measurement: The cost function $E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|H|\psi(\boldsymbol{\theta})\rangle$ is estimated by Hamiltonian averaging using finite-shot measurements. To reduce measurement overhead, commuting Pauli terms can be grouped and measured simultaneously.
Optimization: The parameters $\boldsymbol{\theta}$ of the ansatz are updated iteratively using a classical optimizer (e.g., COBYLA, Adam, SPSA or SLSQP) until the energy converges.

In the following, we provide a more detailed explanation of our choice of encoding and ansatz tailored to the nuclear lattice model considered in this work.

A. Encoding

Implementing quantum algorithms for nuclear Hamiltonian eigenproblems necessitates a well-defined encoding to map fermionic basis states and operators onto their qubit counterparts. The choice of encoding strategy critically governs the computational resource overhead, specifically the qubit count, circuit depth, and gate complexity. In this study, we evaluate two distinct schemes: the canonical Jordan–Wigner transformation and the Gray code encoding. The two approaches differ fundamentally in their utilization of physical symmetries and the efficiency with which they represent the relevant Hilbert space.

1. *Jordan–Wigner Transformation*

Among the available fermion-to-qubit encoding schemes, the Jordan–Wigner (JW) transformation is the most straightforward and widely adopted. To account for the four spin-isospin degrees of freedom inherent to

nuclear systems, the qubit register is partitioned into four sectors, each corresponding to a distinct spin-isospin state. Each sector comprises L^3 qubits, matching the number of lattice sites. Within each block, the qubit state $|1\rangle_j$ ($|0\rangle_j$) signifies that the j-th lattice site is occupied (unoccupied) by a fermion. The fermionic creation and annihilation operators are mapped to qubit operators as follows:

$$\hat{c}_j^\dagger = \frac{1}{2}(X_j - iY_j) \otimes \bigotimes_{k<j} Z_k, \tag{4}$$

$$\hat{c}_j = \frac{1}{2}(X_j + iY_j) \otimes \bigotimes_{k<j} Z_k, \tag{5}$$

where the nonlocal string of Z operators preserves the fermionic anti-commutation relations and ensures the correct anti-symmetry of the wave function.

For the Hamiltonian in Eq. (2), the number of hopping terms scales as $\mathcal{O}(L^6)$, while the on-site multi-body terms scale as $\mathcal{O}(L^3)$. After the JW mapping, the number of Pauli terms is still dominated by the hopping part and scales as $\mathcal{O}(L^6)$ up to a constant factor.

While the JW encoding is conceptually straightforward, it poses significant challenges for noisy intermediate-scale quantum (NISQ) devices. Achieving high-precision results typically requires a large lattice size L, resulting in qubit overheads that exceed the capacity of current hardware. Moreover, the physically relevant Hilbert subspace—constrained by particle number conservation and other symmetries—is exponentially smaller than the full qubit space spanned by the JW encoding. This discrepancy motivates the adoption of compact encoding schemes (for example, the Gray code encoding introduced in the next section) that can exploit symmetries to truncate the Hilbert space. In the era of fault-tolerant quantum computing, the JW encoding may regain prominence; its resource requirements scale linearly with the lattice volume rather than the particle number, offering a scalable pathway once qubit limitations are alleviated.

2. *Gray Code Encoding*

To minimize qubit overhead, we first construct a symmetry-adapted basis that respects all relevant symmetries, such as particle-number conservation and momentum conservation (as detailed in the subsequent section). This procedure constrains the problem to a reduced Hilbert subspace of dimension N.

Subsequently, we employ the Gray code encoding to map each physical basis state $|k\rangle$ in this subspace to a multi-qubit state $|g_k\rangle$, where $g_k \in G_\eta$. Here, $\eta = \lceil \log_2 N \rceil$ represents the minimum number of qubits required, and G_η denotes the set of η-bit binary strings ordered such that adjacent elements have a Hamming distance of exactly one. This dense encoding scheme effectively utilizes the qubit Hilbert space, resulting in an exponential reduction in qubit requirements compared to the Jordan–Wigner encoding.

For instance, consider a reduced subspace of dimension $N = 8$; this requires only $\eta = \log_2(8) = 3$ qubits. The corresponding Gray code mapping is given by:

$$|0\rangle \mapsto |000\rangle, \quad |1\rangle \mapsto |001\rangle, \quad |2\rangle \mapsto |011\rangle, \quad |3\rangle \mapsto |010\rangle,$$
$$|4\rangle \mapsto |110\rangle, \quad |5\rangle \mapsto |111\rangle, \quad |6\rangle \mapsto |101\rangle, \quad |7\rangle \mapsto |100\rangle.$$

Due to the lack of a direct physical interpretation for the Gray code basis states, the fermionic creation and annihilation operators cannot be straightforwardly mapped to compact local qubit operations. Consequently, the qubit Hamiltonian H_q is constructed by explicitly projecting the Hamiltonian matrix elements onto the encoded basis states:

$$H_q = \sum_{i,j=0}^{N-1} \langle i|H|j\rangle \, |g_i\rangle \langle g_j|, \tag{6}$$

where N denotes the dimension of the reduced Hilbert space. The outer products $|g_i\rangle \langle g_j|$ are then decomposed into tensor products of single-qubit operators using the following relations:

$$|0\rangle\langle 1| = \sigma^+ = \frac{1}{2}(X + iY), \quad |1\rangle\langle 0| = \sigma^- = \frac{1}{2}(X - iY),$$
$$|0\rangle\langle 0| = \frac{1}{2}(I + Z), \quad |1\rangle\langle 1| = \frac{1}{2}(I - Z).$$

As the qubit Hamiltonian is built from matrix elements in the reduced basis, after expanding the operators into Pauli strings, the number of terms can grow quickly with the reduced dimension N; in the worst case it scales as $\mathcal{O}(N^3)$ (up to cancellations). For this reason, Gray code encoding is mainly useful when the reduced Hilbert space remains small, which in our setting corresponds to the low-particle-number regime. Check!

B. Hardware-Efficient Ansatz

Since the Gray code basis lacks a direct physical interpretation, it prevents the straightforward application of physically motivated ansatzes such as unitary coupled cluster (UCC). However, because the encoding itself enforces all relevant symmetries, there is no need to impose additional symmetry constraints at the circuit level. This significantly reduces the required gate overhead. For these reasons, we employ the hardware-efficient ansatz (HEA) in this work. The HEA consists of repeated layers of parameterized single-qubit rotations and two-qubit entangling gates. The structure makes it well suited for near-term quantum devices.

Since the ground-state wave function of the nuclear Hamiltonian is real-valued, we adopt a simplified ansatz in which the single-qubit gates are restricted to R_y gates, which is sufficient to represent real amplitudes while keeping the circuit depth minimal. A schematic diagram of a 3-qubit ansatz is shown in Fig. 1.

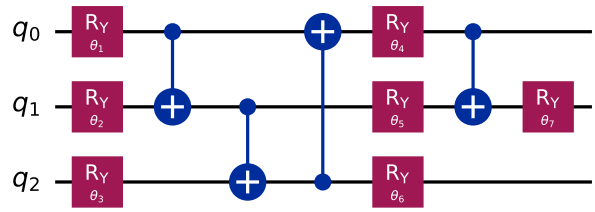

FIG. 1. Schematic diagram of a 3-qubit streamlined hardware-efficient ansatz.

IV. QUANTUM RESOURCES REDUCTION USING SYMMETRIES

A. Dimension of the Symmetrized Hilbert Space

We start from the nuclear Hamiltonian defined in Eq. (2). The model is formulated on a three-dimensional cubic lattice with periodic boundary conditions. At each lattice site there is a single spatial orbital and four internal degrees of freedom corresponding to spin and isospin. Therefore, each site can host at most four fermions due to the Pauli principle.

Throughout this work, we consider a fixed total particle number $n < 4L^3$. The full Fock-space dimension at fixed particle number is given by the number of ways to distribute n fermions among the available single-particle states. Since the total number of single-particle states is $4L^3$, the dimension of the Hilbert space is

$$\dim(\mathcal{H}) = \binom{4L^3}{n}. \tag{7}$$

If proton and neutron numbers are fixed separately, N_p and N_n, then the dimension becomes

$$\dim(\mathcal{H}) = \binom{2L^3}{N_p}\binom{2L^3}{N_n}. \tag{8}$$

We now use the spatial symmetry group G to construct projection operators and restrict the Hilbert space to the subspace consistent with the symmetry of the ground state. For a finite cubic lattice with periodic boundary conditions, this symmetry group G is given by the semidirect product of the discrete translation group T and the cubic point group O_h:

$$G = T \rtimes O_h, \tag{9}$$

where T is the finite Abelian group formed by L^3 discrete translations, and O_h denotes the full cubic point group comprising 48 elements (including rotations, reflections, and spatial inversion). In practice, we treat the translation and point group symmetries separately.

First, for a translationally invariant Hamiltonian, the ground state lies in the zero-momentum sector ($\mathbf{k} = \mathbf{0}$). we therefore use the translation group T to construct the projection operator onto this subspace:

$$P_{k=0} = \frac{1}{|T|}\sum_{\mathbf{t}\in T}\hat{T}_{\mathbf{t}}, \tag{10}$$

where $|T| = L^3$ is the number of lattice translations. Given an arbitrary Fock state $|\psi\rangle$ with fixed particle number, a corresponding zero-momentum state can be constructed as

$$|\Psi_{k=0}\rangle = \frac{1}{\sqrt{|T|}}\sum_{\mathbf{t}\in T}\hat{T}_{\mathbf{t}}\,|\psi\rangle\,. \tag{11}$$

In general, by exploiting translational invariance, we reduces the Hilbert space dimension approximately by a factor of L^3, up to corrections from configurations that are invariant under certain translations.

Next, we reduce the Hilbert space dimension by using point group symmetries. In the zero-momentum sector ($\mathbf{k} = \mathbf{0}$), the wave vector is invariant under any point group operation R (i.e., $R\,\mathbf{0} = \mathbf{0}$). Consequently, the associated little group $G_{\mathbf{k}}$ coincides with the full cubic point group O_h. For a translationally invariant system without spontaneous symmetry breaking, a non-degenerate ground state transforms according to the trivial irreducible representation of the lattice symmetry group. In the cubic group O_h, this representation is A_{1g}, which corresponds to a lattice scalar. The corresponding projection operator is thus defined as:

$$\hat{P}_{A_{1g}} = \frac{1}{|O_h|}\sum_{R\in O_h}\chi^{(A_{1g})}(R)^*\hat{R} = \frac{1}{48}\sum_{R\in O_h}\hat{R}, \tag{12}$$

where the character $\chi^{(A_{1g})}(R) = 1$ for all $R \in O_h$. The symmetry-adapted basis state $|\Phi_{\rm sy}\rangle$ is explicitly derived as:

$$|\Phi_{\rm sy}\rangle = \hat{P}_{A_{1g}}\,|\Psi_{k=0}\rangle = \frac{1}{48\sqrt{|T|}}\sum_{R\in O_h}\sum_{\mathbf{t}\in T}\hat{R}\hat{T}_{\mathbf{t}}\,|\psi\rangle\,. \tag{13}$$

Ultimately, the cumulative application of these spatial symmetries yields a substantial reduction of the size of the Hilbert space relevant for the ground state. Starting from the Fock space with fixed particle number n, whose dimension is $\binom{4L^3}{n}$, projection onto the zero-momentum and A_{1g} sectors reduces the dimension by approximately a factor of $48L^3$. Consequently, the dimension of the resulting symmetry-reduced Hilbert space can be estimated by:

$$\dim(\mathcal{H}) \approx \frac{1}{48L^3}\binom{4L^3}{n}. \tag{14}$$

In this work, we consider $n \leq 4$ and $L \in \{2, 3, 4, 5, 6\}$. In the regime where $L^3 \gg n$, the binomial coefficient can be

approximated as:

$$\binom{4L^3}{n} \approx \frac{(4L^3)^n}{n!}. \tag{15}$$

which leads to:

$$\dim(\mathcal{H}) \approx \frac{4^n (L^3)^{n-1}}{48 n!}. \tag{16}$$

If proton and neutron numbers are fixed separately, the Fock-space dimension becomes $\binom{2L^3}{N_p}\binom{2L^3}{N_n}$. After imposing translation and point-group symmetries, the reduced dimension is approximately:

$$\dim(\mathcal{H}) \approx \frac{2^n (L^3)^{n-1}}{48 N_p! N_n!}. \tag{17}$$

Has spin conservation been considered here? The estimation needs to be close to the values in Table 1.

B. Symmetry-Based Qubit Reduction

In this section, we use Gray code encoding to map the symmetry-adapted basis states. As discussed above, encoding an N-dimensional Hilbert space requires a qubit register size of $\lceil \log_2 N \rceil$. Consequently, the required number of qubits, denoted as N_q, is approximated by:

$$N_q \approx 2n + (n-1)\log_2(L^3) - \log_2 n! - \log_2 48. \tag{18}$$

For comparison, the JW encoding maps each lattice site and internal degree of freedom directly to qubits. In that case, the required number of qubits scales as $N_q^{\mathrm{JW}} = 4L^3$. Note that one may taper a few qubits using $\mathbb{Z}_2$ parity symmetries associated with conserved particle numbers (and, when applicable, spin-resolved particle numbers). This reduces the register by a constant number of qubits but does not change the $\mathcal{O}(L^3)$ scaling.

Table I compares the Gray code encoding with the JW encoding for different lattice sizes L. In this calculation, we fix $N_p = 1$, $N_n = 2$ and total spin $S = 1/2$ (??), in addition to the spatial symmetry constraints discussed above. The qubit numbers reported in the table are obtained under these symmetry restrictions.

The JW encoding scales with the lattice volume L^3, while the Gray code encoding scales approximately with $\log_2(\mathrm{L}^3)$ after symmetry reduction. As L increases from 2 to 6, the qubit number required by JW grows from 24 to 648. In contrast, the Gray code encoding requires 9 qubits at $L = 6$.

This comparison shows that symmetry reduction combined with Gray code encoding significantly lowers the qubit requirement in the low-particle-number regime considered here. However, since the encoded basis no longer has a direct physical interpretation, physics-inspired ansatz constructions are not readily available, and the resulting variational circuits typically require greater depth for a fixed qubit number.

TABLE I. Comparison of the qubit count for Gray Code (GC) and Jordan–Wigner (JW) encodings as a function of the linear dimension L of the three-dimensional lattice, at a particle number $n = 3$.

Linear Dimension(L)	Qubit Count (N_q)	
	Gray Code	Jordan–Wigner
2	3	24
3	4	81
4	6	192
5	7	375
6	9	648

V. NUMERICAL SIMULATIONS

In this section, we present a numerical investigation of the ground-state properties of light nuclei, including deuteron(^{2}H), triton(^{3}H), and helium-4(^{4}He) using VQE. The nuclear systems are modeled on finite lattices with linear dimension ranging from $L = 2$ to $L = 6$. Due to the prohibitive qubit overhead associated with the standard Jordan-Wigner transformation at these lattice sizes, we exclusively employ the resource-efficient Gray code encoding to map the nuclear states onto the qubit computational basis. All variational protocols are emulated classically using the PennyLane framework.Add citation Gradients are computed efficiently with JAX, and the parameters are optimized using the Adam optimizer in Optax.

Figure 2 shows how the ground-state binding energies depend on L for ^{2}H, ^{3}H, and ^{4}He. Our goal here is to see how the finite volume affects the results when L is changed. Is this our goal?

We first fix the interaction parameters at $L = 6$. At this lattice size, we fit the two-body coupling strength c_2 in Eq. (2) so that the calculated ground-state energy of ^{2}H matches the experimental binding energy. We then fit the three-body coupling c_3 using the experimental ground-state energy of ^{3}H. After this fit, we keep the same values of (c_2, c_3) for all other lattice sizes. With these fitted parameters, we compute the ground-state energy of ^{4}He as a check. As shown in Fig. 2c, the ^{4}He result at $L = 6$ is also very close to the experimental value. In practice, if heavier systems are considered, the same fitted (c_2, c_3) will be used. Is this description ok?

When a smaller L is adopted, the results deviate away from the experimental binding energies for all three nuclei. The deviation is largest at small L. This behavior is mainly a finite-volume effect under periodic boundary conditions: when the lattice is small, the wave function overlaps more strongly with its periodic images, and the long-range part of the bound-state wave function is distorted by the limited lattice size. As L increases, these effects become smaller, and the energies approach the experimental benchmarks. Is this explanation OK?

Figure 3 shows the convergence behavior of the VQE

for deuteron ($n = 2$), triton ($n = 3$), and ^{4}He ($n = 4$). We plot the energy residual ΔE as a function of iteration and set the stopping accuracy of 10^{-6} MeV in our simulations.

The number of iterations needed to reach this level depends on n and L. When n increases, convergence becomes slower. At fixed n, increasing L also makes the optimization harder. This trend is consistent with the fact that the effective Hilbert space grows with n and L, so the optimization has to search a larger parameter space.

VI. CONCLUSION

ACKNOWLEDGMENTS

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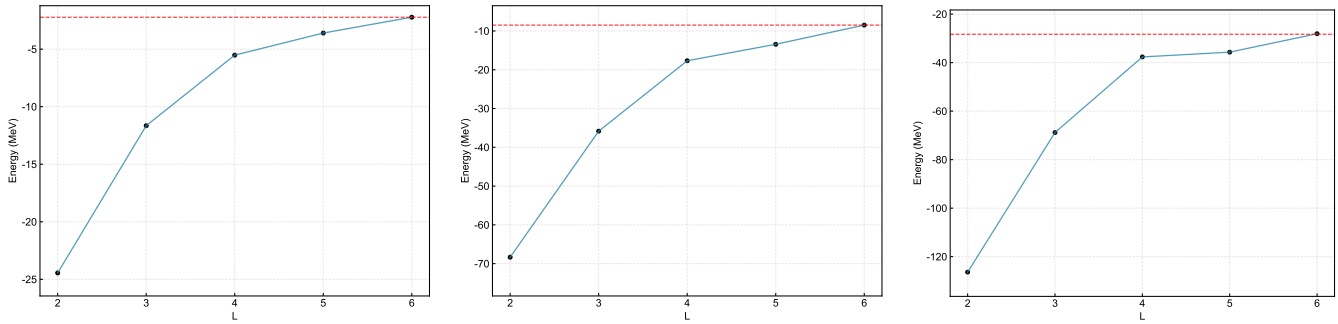
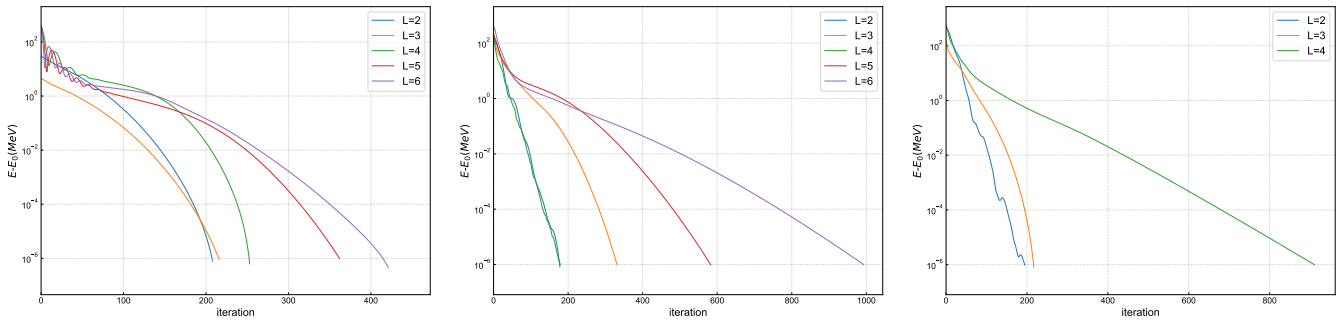


(a) Deuteron (^{2}H) (b) Triton (^{3}H) (c) Helium-4 (^{4}He)

FIG. 2. This figure needs to be regenerated Ground-state binding energies versus lattice size L. Each data point (black circle) represents the variational energy minimum achieved via VQE optimization for the corresponding lattice dimension L and nucleon number n. Panels display results for (a) the deuteron (^{2}H, $n = 2$), (b) the triton (^{3}H, $n = 3$), and (c) helium-4 (^{4}He, $n = 4$). Red dashed lines indicate the experimental binding energy benchmarks ($E_{\text{exp}}^{^2\text{H}} \approx -2.22$ MeV, $E_{\text{exp}}^{^3\text{H}} \approx -8.48$ MeV, and $E_{\text{exp}}^{^4\text{He}} \approx -28.30$ MeV).



(a) Deuteron (^{2}H) (b) Triton(^{3}H) (c) Helium-4(^{4}He)

FIG. 3. This figure needs to be replotted Convergence profiles of the variational ground-state energy for light nuclei: (a) deuteron (^{2}H, $n = 2$), (b) triton (^{3}H, $n = 3$), and (c) helium-4 (^{4}He, $n = 4$). The energy residual, defined as $\Delta E(\theta_t) = E(\theta_t) - E_0$ (where E_0 is the exact-diagonalization energy for the same Hamiltonian) Specify ΔE, is plotted on a semi-logarithmic scale against the optimization iteration t for various lattice dimensions L.